

CONSTRAINT-BASED PROBLEM SOLVING

CSA07 – Computer Networks | CO5 Assessment Tool 1

Assessment Weightage: 40% | Total Questions: 3 | Total Marks: 30

Course Outcome: CO5 – Measure network performance using key metrics with simulation tools. The assessment requires network solutions to be designed by identifying constraints, decomposing problems, applying constraints, using logical reasoning, and producing feasible solutions.

Purpose of the Work

This document develops structured solutions for the three constraint-based networking problems given in the assessment. Each solution considers the major practical constraints stated in the question: bandwidth, reliability, cost, scalability, coverage, interference, security, performance, compliance, and operational feasibility.

Assessment Problems

Problem	Scenario	Main Design Focus
1	Remote rural branch office	Connectivity, reliability, security, cost, scalability
2	University campus with 3,000+ students	Wireless coverage, capacity, interference, authentication, cost
3	Financial institution	Segmentation, firewall, authentication, IDS, compliance, performance

Important Approach

1. Understand the problem and explicitly identify every constraint.
2. Break the complete problem into smaller network-design sub-problems.
3. Select technologies and devices that satisfy the constraints.
4. Compare performance, reliability, security and cost before finalizing the design.
5. Explain why the selected solution is feasible and scalable.

The assessment rubric gives the highest emphasis to understanding constraints (20%) and applying constraints in the solution (20%). Therefore, these two areas are addressed explicitly in every problem. The other rubric areas—decomposition, reasoning, feasibility, creativity and presentation—are also incorporated throughout the document.

1. Remote Rural Branch Office – Constraint-Based Network Design

Assessment requirement: A multinational company is opening a branch office in a remote rural location where high-speed wired internet is unavailable. The office needs cloud applications, video conferencing and secure communication with headquarters, while operating under a limited budget and requiring uninterrupted business operations. □filecite□turn0file0□L14-L21□

1.1 Understanding the Problem and Constraints

Constraint	Requirement / Impact
Availability	Wired high-speed Internet is unavailable; an alternative access medium is required.
Bandwidth	Cloud applications and video conferencing require stable usable bandwidth.
Reliability	Business operations should continue even when the primary link fails.
Security	Communication with headquarters and cloud services must be protected.
Cost	The solution must fit a limited connectivity and equipment budget.
Scalability	The network should support future users, devices and higher traffic.
Performance	Latency, packet loss, throughput and jitter should remain acceptable.

1.2 Recommended Connectivity

A practical primary choice is a fixed wireless or 4G/5G cellular broadband connection, depending on availability at the rural site. If terrestrial wireless service is not reliable or unavailable, satellite Internet can be considered as the alternative. A dual-WAN design is recommended so that a second connection can take over when the primary connection becomes unavailable.

- Primary WAN: fixed wireless or 4G/5G broadband where reliable signal coverage exists.
- Backup WAN: a second independent cellular provider or satellite link where practical.
- Use automatic WAN failover to maintain business continuity.
- Apply traffic prioritization so voice/video and essential cloud traffic receive suitable priority.
- Choose the access technology based on actual signal strength, data limits, latency and monthly cost.

1.3 Recommended Networking Devices

Device	Purpose
Business-grade router	WAN termination, routing, NAT, VPN and failover
Dual-WAN/SD-WAN capable gateway	Automatic switching and link management
Managed switch	Connects office computers, printers, access points and other devices
Wireless access point	Provides secure Wi-Fi for employees
UPS	Keeps network equipment operating during short power interruptions
VPN gateway/function	Provides encrypted communication with headquarters

1.4 Security Mechanisms and Network Architecture

The branch should use a layered security model. The gateway should provide stateful firewall filtering, secure VPN connectivity to headquarters, strong administrative passwords, firmware updates and logging. Employee Wi-Fi should use enterprise-grade authentication where supported, while a separate guest network should prevent visitors from accessing internal systems.

- Site-to-site VPN: encrypts branch-to-headquarters traffic over the Internet.
- Firewall: blocks unauthorized inbound connections and controls outbound traffic.
- Network segmentation: separates employee devices, guest devices, printers/IoT and management traffic.
- Secure Wi-Fi: use strong encryption and authentication; avoid open internal Wi-Fi.
- Access control: provide only the permissions required for each user or device.
- Monitoring: record link status, bandwidth use, unusual traffic and security events.
- Backup configuration: retain a secure copy of router/firewall settings for quick recovery.

1.5 Constraint Application and Reasoning

Constraint	Design Decision	Why It Fits
Limited wired availability	Wireless/cellular/satellite WAN	Avoids dependence on unavailable wired broadband.
Uninterrupted operation	Dual WAN + automatic failover	A failed primary link does not immediately stop business traffic.
Video conferencing	QoS/traffic prioritization	Reduces the effect of congestion on real-time traffic.
Secure HQ access	Site-to-site VPN + firewall	Protects business communication across an untrusted WAN.
Limited budget	Use scalable business gateway and phased upgrades	Controls initial cost while preserving expansion options.
Future growth	Managed switch + scalable AP and WAN design	Allows additional users/devices without redesigning the whole network.

1.6 Performance Metrics to Monitor

- Throughput: actual data rate available to users.
- Latency: time required for packets to travel between endpoints.
- Jitter: variation in packet delay; important for video conferencing.
- Packet loss: percentage of packets that do not reach the destination.
- Availability: percentage of time the WAN and essential services remain operational.
- Bandwidth utilization: helps identify congestion and plan upgrades.

For the assessment's CO5 focus, the network can be evaluated with simulated or measured traffic by comparing throughput, delay, packet loss and availability under normal and failover conditions.

2. University Campus Wireless Network for 3,000+ Students

Assessment requirement: The university must provide campus-wide wireless access for more than 3,000 students across classrooms, laboratories, libraries and hostels. The design must support many simultaneous users, minimize interference, provide secure access and work within a limited access-point budget.

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2.1 Understanding the Requirements and Constraints

Constraint	Design Implication
Large user population	Capacity planning is as important as simple coverage.
Simultaneous usage	High-density areas require careful AP placement and channel planning.
Limited AP count	APs must be positioned where demand is highest; unnecessary overlap must be avoided.
Interference	Use systematic channel planning and appropriate transmit power.
Security	Students and staff must authenticate before accessing protected resources.
Coverage	Classrooms, labs, library and hostel zones require appropriate signal coverage.
Cost	Prefer efficient deployment and centralized management rather than excessive hardware.

2.2 Proposed Wireless Architecture

A centrally managed Wi-Fi architecture is recommended. Access points should connect through managed PoE switches to a core/distribution network. A wireless controller or cloud-managed platform can simplify configuration, monitoring, firmware management and roaming policies. The design should be capacity-based rather than simply placing one AP wherever a coverage gap appears.

2.3 Access Point Placement

- Perform a site survey before final placement to identify walls, interference sources and high-density areas.
- Place APs near classrooms, laboratories, libraries and other high-demand locations.
- Use more AP capacity in lecture halls and crowded common areas rather than distributing APs equally by building area.
- Use controlled cell overlap to support roaming without creating excessive co-channel interference.
- Mount APs in suitable positions away from major physical obstructions and interference sources.
- Use the limited AP budget where simultaneous user density is highest.

2.4 Frequency Band Selection

Use the 5 GHz band as the main high-capacity band where client support is available because it generally provides more usable channels and is better suited to high-density deployments. The 2.4 GHz band can be retained for legacy devices and longer-range coverage but should be carefully planned because it has fewer non-overlapping channels and is more prone to congestion.

Band	Recommended Role	Key Point
2.4 GHz	Legacy/low-bandwidth devices and selected coverage areas	Longer reach but more congestion
5 GHz	Primary high-capacity student access	Better choice for dense high-throughput usage
6 GHz (if supported)	Future expansion / compatible modern clients	Can add capacity where equipment and regulations support it

2.5 Authentication and Security

- Use enterprise authentication for students and staff rather than a shared password.
- Integrate authentication with the university identity system where feasible.
- Use separate logical networks/VLANs for students, staff, guests and infrastructure.
- Apply access-control policies so users can reach only the resources required for their role.
- Use secure management access for APs, switches and controllers.
- Monitor rogue APs, abnormal traffic, repeated authentication failures and other suspicious events.

2.6 Balancing Coverage, Performance, Security and Cost

Goal	Method
Coverage	Site survey + strategic AP placement
Performance	Capacity planning + 5 GHz preference + channel planning
Interference reduction	Non-overlapping channel allocation + transmit-power tuning
Security	Enterprise authentication + VLAN separation + access policies
Cost control	Deploy APs according to user density and traffic demand
Scalability	Centralized management and expandable switching/backhaul

2.7 Performance Evaluation

The network should be tested in representative locations and at busy times. Important measurements include throughput per user, average latency, packet loss, signal quality, channel utilization, number of connected clients per AP and roaming performance.

- Measure throughput during normal and peak usage.
- Measure latency and packet loss for real-time applications.
- Check whether any AP becomes overloaded while nearby APs remain underused.
- Review channel utilization and interference before increasing transmit power.
- Use monitoring data to decide whether additional capacity is needed in specific high-density areas.

2.8 Recommended Logical Design

Internet/ISP → Edge Router/Firewall → Core Switch → Managed PoE Access Switches → Campus Access Points → Student/Staff/Guest Wireless Networks. Authentication services and monitoring systems should be connected through protected network segments. This structure separates wireless access from the core network and makes policy enforcement easier.

This design directly addresses the assessment's required balance between coverage, performance, security and cost. □filecite□turn0file0□L25-L29□

3. Financial Institution – Secure Network Architecture

Assessment requirement: A financial institution is upgrading its internal network to strengthen cybersecurity while employees still need access to sensitive customer information and online banking services. The organization must meet strict security policies without significantly reducing performance or increasing operational cost.

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3.1 Security and Operational Constraints

Constraint	Requirement
Cybersecurity	Prevent unauthorized access and reduce attack impact.
Sensitive data	Customer and financial information requires strong access control.
Performance	Security controls should introduce minimal unnecessary delay.
Compliance	Security policies, logging and access controls must be enforceable and auditable.
Cost	Reuse scalable infrastructure and avoid unnecessary duplication.
Availability	Critical banking and internal services should remain accessible to authorized users.
Manageability	Security policies should be centrally monitored and maintained.

3.2 Network Segmentation

Network segmentation is a primary control because it limits lateral movement if one device or account is compromised. The institution should logically separate departments and security zones based on function and sensitivity.

- User LAN: normal employee workstations.
- Sensitive-data zone: systems handling confidential customer information.
- Server zone: application and database servers with tightly controlled access.
- Guest network: Internet-only access, isolated from internal resources.
- Management network: restricted access for administrators and network devices.
- DMZ: public-facing services that must be reachable from outside but should not sit directly inside the trusted internal network.

3.3 Firewall Deployment

Place a next-generation firewall at the Internet edge and use internal segmentation firewalls or equivalent controls between high-risk zones when justified by the architecture. Firewall rules should follow a default-deny approach for unnecessary traffic and explicitly permit required services. Logging should be enabled for important security events and policy violations.

3.4 Authentication Methods

- Use individual user accounts instead of shared credentials.
- Use multi-factor authentication for sensitive systems and administrative access.
- Apply role-based access control so employees receive only necessary permissions.
- Use strong password policies and account lockout/risk controls where appropriate.
- Review privileged accounts regularly and remove unnecessary access.

3.5 Intrusion Detection and Monitoring

Intrusion Detection Systems (IDS) and, where appropriate, Intrusion Prevention Systems (IPS) can monitor network traffic for suspicious patterns. Security monitoring should combine firewall logs, authentication events, endpoint alerts and network telemetry so that unusual activity can be identified quickly.

- Network IDS/IPS: identifies suspicious network traffic and known attack patterns.
- Security log collection: centralizes important events for investigation.
- Alerting: notifies administrators about high-risk events.
- Traffic monitoring: identifies unusual traffic volume or destinations.
- Periodic review: verifies that rules, accounts and segmentation remain effective.

3.6 Security Architecture

A suitable high-level architecture is: Internet → Edge Router → Next-Generation Firewall → DMZ / Internal Core → Segmented VLANs and Security Zones → Servers and User Networks. Administrative access should use a restricted management network. Monitoring and logging should collect security events from the firewall, authentication system, servers and network infrastructure.

3.7 Applying the Constraints

Constraint	Solution	Expected Benefit
High security	Segmentation + firewall + MFA + IDS/IPS	Reduces unauthorized access and limits attack spread.
Sensitive customer information	Dedicated protected zone + role-based access	Restricts access to authorized users and systems.
Performance	Rule optimization + capacity planning + targeted inspection	Avoids unnecessary processing and congestion.
Compliance	Central logging + access reviews + documented policies	Supports auditing and security governance.
Budget	Prioritize high-risk zones and centrally manage controls	Provides strong security without uncontrolled hardware growth.

3.8 Feasibility and Performance

The architecture is feasible because it uses standard enterprise networking concepts and can be implemented in stages. First, segmentation and firewall policy can be established; next, stronger authentication and monitoring can be introduced; finally, advanced detection and optimization can be added. Performance should be monitored using latency, throughput, packet loss, firewall utilization and application response time.

4. Rubric-Oriented Final Analysis

The assessment rubric evaluates seven criteria: understanding of the problem and constraints (20%), problem decomposition (15%), application of constraints in the solution (20%), logical reasoning and strategy (15%), solution accuracy and feasibility (15%), creativity and innovation (5%), and clarity of presentation (10%).

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Rubric Criterion	How This Work Addresses It
Understanding of Problem & Constraints – 20%	Every problem begins with explicit constraints such as cost, bandwidth, reliability, interference, security, performance and compliance.
Problem Decomposition – 15%	Each scenario is divided into connectivity, devices, security, architecture, performance and evaluation sub-problems.
Application of Constraints – 20%	Technology choices are directly connected to the constraints rather than being listed without justification.
Logical Reasoning & Strategy – 15%	Alternatives are compared and the recommended architecture is selected based on practical trade-offs.
Solution Accuracy & Feasibility – 15%	The designs use standard network concepts and can be implemented in stages.
Creativity & Innovation – 5%	Examples include dual-WAN failover, capacity-based Wi-Fi deployment and layered security zones.
Clarity of Presentation – 10%	Tables, headings, structured explanations and a final comparison make the solutions easy to follow.

5. Key Points to Remember for Viva / Presentation

6. Constraint-based problem solving means designing a solution while respecting real-world limitations.
7. Do not select a network technology only because it is fast; consider cost, reliability, security and scalability.
8. For a rural branch, redundancy and secure VPN connectivity are major priorities.
9. For a university with 3,000+ users, capacity and interference management are as important as coverage.
10. For a financial institution, segmentation limits the impact of a security breach.
11. Firewall rules should permit required traffic and block unnecessary access.
12. Authentication verifies users; authorization determines what they are allowed to access.
13. IDS/IPS and centralized monitoring help detect suspicious activity.
14. Important network performance metrics include throughput, latency, jitter, packet loss and availability.
15. Simulation or measurement should compare network performance under realistic traffic and failure conditions.

6. Conclusion

The three scenarios demonstrate that effective computer-network design is a constraint-based decision process. The best solution is not necessarily the most expensive or highest-performance option; it is the solution that satisfies the required performance, security, reliability and scalability within the stated constraints. For CO5, the proposed designs can be evaluated using measurable network metrics such as throughput, latency, packet loss, jitter, utilization and availability. This makes the solutions suitable for both practical network planning and performance-oriented assessment.

Source basis: Assessment Tool 1, CSA07 – Computer Networks, CO5, Constraint-Based Problem Solving. The three problem statements and rubric criteria are taken from the uploaded assessment document.

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